

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCM1363 DISCRETE MATHEMATICS

THURSDAY, 18 MAY 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

Answer ALL questions in **section A**.

Answer ANY ONE question in **section B**.

If more than ONE question from **section B** is answered, then only the **first question attempted in section B will be marked**.

All questions carry equal marks. Marks allocated for each part of the questions are indicated in brackets.

UCCM1363 DISCRETE MATHEMATICS**SECTION A**Answer ALL questions.

- Q1. (a) Write the logical form of each argument and use a truth table to determine the validity of the arguments. (10 marks)

If coconut is not in basket 1, then guava is in basket 2.

If guava is not in basket 2, then coconut is in basket 1.

Therefore, coconut is not in basket 1 or guava is not in basket 2.

- (b) For each of the following statement, rewrite the statement in English without using the symbol $\forall$ or $\exists$ or variables. Then write a negation for the statements.

(i) $\forall$ odd integers n , $\exists$ an integer k such that $n = 2k + 1$. (5 marks)

(ii) $\forall x \in \mathbb{R}$, $\exists$ a real number y such that $x + y = 0$. (5 marks)

- (c) Without using a truth table, show that $\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$. (5 marks)
[Total : 25 marks]

- Q2. (a) Obtain the explicit formula for the following sequence. (10 marks)

$$\frac{7}{2}, 11, \frac{45}{2}, 38, \frac{115}{2}, \dots$$

- (b) Prove that $4^n - 2^n$ is divisible by 2 for $n \geq 1$, by using mathematical induction. (8 marks)

- (c) Use iteration to guess an explicit formula $S_0 = 1$ and $S_k = 4S_{k-1} + 3$, $k \geq 1$. (7 marks)
[Total : 25 marks]

- Q3. (a) Determine whether the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 5\}$ is a lattice. (5 marks)

- (b) Explain how to identify whether a lattice is a distributive lattice by giving a simple example. (8 marks)

UCCM1363 DISCRETE MATHEMATICS**Q3. (Continued)**

- (c) Let $A = \{1, 2, 3, 4\}$. For the relation R whose matrix is given as follow, find the matrix of the transitive closure by using Warshall's algorithm. (12 marks)

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

[Total : 25 marks]

SECTION BAnswer only ONE question.

- Q4. (a) State the requirements or criteria to be a tree. (5 marks)

- (b) Application of algorithm for the *inorder* and *postorder* traversals of a binary tree with 17 nodes yields the following sequences of nodes.

INORDER : A B C F E D G P I H O J K M L N
 POSTORDER: A B C D E F G H I J K L M N O P

Construct the diagram of the binary tree. (12 marks)

- (c) Construct a label tree represents the following algebraic expression. (8 marks)

$$(8 \times (7 \div (6 - 5))) \times ((4 \div (3 \times 2)) - 1)$$

[Total : 25 marks]

- Q5. (a) Construct a label tree represents the following algebraic expression. (8 marks)

$$(4 \times (4 \div (8 - 2))) \div ((7 + (6 \div 3)) + 9)$$

- (b) If T and T' are two spanning trees in G , is it necessary that T and T' have an edge in common? Either prove this or produce a counterexample. (6 marks)

- (c) Form a binary search tree to represent $\{4, 6, 2, 8, 5, 3, 7, 1\}$. (6 marks)

- (d) What is meant by adversarial search, and how does it differ from traditional tree search. (5 marks)

[Total : 25 marks]

UCCM1363 DISCRETE MATHEMATICS**Formula Sheet**

Let p and q be statement variables: (1) The negation of $p \Rightarrow q$ is $p \wedge \sim q$. (2) The contrapositive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$. (3) The converse of $p \Rightarrow q$ is $q \Rightarrow p$. (4) The inverse of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$. Theorem 1.1 Let p and q be statements variables. (1) $p \Rightarrow q \equiv \sim p \vee q$ (2) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$ (3) $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$ Theorem 1.2 (1) Commutative laws: $p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$ (2) Associative laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$ (3) Distributive laws: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (4) Identity laws: $p \wedge t \equiv p$ $p \vee c \equiv p$ (5) Negation laws: $p \vee \sim p \equiv t$ $p \wedge \sim p \equiv c$ (6) Double negative laws: $\sim(\sim p) \equiv p$ (7) Idempotent laws: $p \wedge p \equiv p$ $p \vee p \equiv p$ (8) Universal bound laws: $p \vee t \equiv t$ $p \wedge c \equiv c$ (9) De Morgan's laws: $\sim(p \wedge q) \equiv \sim p \vee \sim q$ $\sim(p \vee q) \equiv \sim p \wedge \sim q$ (10) Absorption laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$ (11) Negations of t and c: $\sim t \equiv c$ $\sim c \equiv t$	Rules of Inference		Name
	1.	$\begin{array}{l} p \Rightarrow q \\ p \\ \hline \therefore q \end{array}$	Modus Ponens (Law of Detachment)
	2.	$\begin{array}{l} p \Rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array}$	Modus Tollens
	3.	$\begin{array}{l} p \quad \text{or} \quad q \\ \hline \therefore p \vee q \end{array}$	Disjunctive Addition
	4.	$\begin{array}{l} p \wedge q \quad \text{or} \quad p \wedge q \\ \hline \therefore p \end{array}$	Conjunctive Simplification
	5.	$\begin{array}{l} p \vee q \quad \text{or} \quad p \vee q \\ \sim p \quad \quad \quad \sim q \\ \hline \therefore q \end{array}$	Disjunctive Syllogism
	6.	$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	Conjunction
	7.	$\begin{array}{l} p \Rightarrow q \\ q \Rightarrow r \\ \hline \therefore p \Rightarrow r \end{array}$	Hypothetical Syllogism
	8.	$\begin{array}{l} p \vee q \\ \sim p \vee r \\ \hline \therefore q \vee r \end{array}$	Resolution